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Review

Mitochondrial Gene Therapy: An Evaluation of Strategies for the Treatment of Mitochondrial DNA Disorders

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Abstract

Mitochondrial DNA (mtDNA) disorders include a vast range of pathological conditions, despite each sharing a mutual inability to produce ATP efficiently as a result of defective oxidative phosphorylation. There is no clear consensus regarding an effective therapeutic approach, and consequently the current treatment strategies are largely supportive rather than curative. This is almost certainly the result of there being virtually no defined genotype–phenotype relationships among the mtDNA disorders; hence an identical mutation may be responsible for multiple phenotypes, or the same phenotype may be produced by different mutations. In light of this, the development of gene therapy to treat mtDNA disorders offers a promising approach, as it potentially circumvents the complication of the aforementioned genotype–phenotype inconsistency and ultimately the current inability to treat individual disorders with sufficient efficacy. Such an approach will ultimately require the combination of efficient mitochondrial targeting, and an effective therapeutic molecule. Although promising proof-of-principle developments in this field have been demonstrated, the realization of a successful therapeutic mitochondrial gene therapy strategy has not come to fruition. This review critiques the key approaches under development by discussing the theory underlying each strategy, and detailing the current progress made. We also emphasize the potential hurdles that must be acknowledged and overcome if the potential of a therapeutic gene therapy to treat mitochondrial DNA disorders is to be realized.

Mitochondrial DNA and Gene Therapy: An Introduction

Mitochondria are cytoplasmic, dual-membrane organelles ubiquitously located within all nucleated cells. Although involved in several central metabolic pathways that are essential for cellular function, mitochondria are predominantly associated with cellular energy production, and hence are often regarded as the powerhouses of the cell. The final reaction of aerobic metabolism, the synthesis of ATP from its precursors ADP and P_i, is catalyzed by F_oF_1-ATP synthase located within the inner mitochondrial membrane, driven by an electrochemical gradient formed by the pumping of protons from the matrix and into the inner membrane space by oxidative phosphorylation (OXPHOS). The oxidative phosphorylation complex, or respiratory chain, which includes F_oF_1-ATP synthase, is composed of five heteromeric complexes, and is the only mammalian cellular pathway under dual genetic control: a complementation of nuclear and mitochondrial encoded genes. Located within the mitochondrial matrix, mitochondrial DNA (mtDNA) encodes 13 polypeptides and 24 RNA transcripts that contribute to 4 of the 5 respiratory chain complexes (Attardi *et al.*, 1982). Although representing less than 1% of total cellular DNA, functional mtDNA is crucial for OXPHOS, and hence essential for cellular function.

mtDNA dysfunction is typically characterized by an inability to meet cellular ATP demand as a result of defective OXPHOS (Taylor *et al.*, 2000). Consequently, it is not surprising that mtDNA dysfunction contributes to a diverse range of human clinical diseases, encompassing both tissue-specific and multiple system disorders, with the onset of disease appearing congenitally or later, even during adult life (Zeviani and Di Donato, 2004). In terms of clinical relevance, mtDNA disorders are among the most prevalent of all genetic disorders, with approximately 1 in 8000 individuals either carrying or affected by a pathogenic mutation (Chinnery *et al.*, 2000; Skladal *et al.*, 2003). Although mtDNA

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disorders are often characterized by obscure but debilitating neurodegenerative and neuromuscular defects, they are also increasingly implicated in a range of socially recognizable conditions, including obesity, diabetes, and cancer (Weissig *et al*., 2004). The extensive phenotypic variability of these disorders currently requires a range of therapeutic strategies to treat defective mtDNA-associated diseases. To date, vitamin supplements, pharmacological agents, and exercise therapy have been investigated; however, a review of such treatment programs concluded there was no clear evidence supporting the effectiveness of any of these approaches (Chinnery *et al*., 2006). There is therefore a clear requirement for new therapeutic approaches.

Genetic manipulation of mtDNA and its gene products is one avenue under investigation for the treatment of mtDNA disorders. As its underlying aim is to correct the defects of OXPHOS itself, the one mutually shared characteristic among mtDNA disorders, it is likely that many of the numerous but ineffective strategies that ultimately treat only the symptoms of individual disorders will become redundant. Thus far, the research undertaken to apply gene therapy for the treatment of mtDNA disorders can be categorized broadly into three approaches. The first aims to discriminate between wild-type and mutant mtDNA, specifically inhibiting mutant mtDNA replication and transcription, and hence preventing mutant gene expression. The second approach involves the delivery of recombinant mtDNA vectors that express functional replacement copies of endogenous defective genes from within the mitochondrial matrix. The third strategy involves the recoding and translocation of mitochondrial genes to be expressed from the nucleus, and their gene products subsequently targeted to mitochondria. Although each of these strategies has their individual merits, in practice their full potential has not been realized (see Table 1 for a summary of the benefits and shortcomings of the three discussed strategies).

Although mitochondrial disorders are caused by both nuclear and mitochondrial DNA defects, this review focuses on the genetic basis of mtDNA disease only, with particular reference to mtDNA point mutations, and discusses the potential for therapeutic genetic intervention to treat such disorders. Foremost, the aim of this review is to appraise each of the three key gene therapy strategies outlined earlier, to present both preceding and current progress made toward the development of the strategies, and finally to provide a critical analysis of the limits and shortcomings of the aforementioned theoretical concepts, for the treatment of mtDNA disorders.

Molecular Pathology and Genetics of mtDNA Disorders

mtDNA disorders encompass a diverse range of conditions that are usually progressive, and are often responsible for severe morbidity and premature mortality (Chinnery *et al*., 2006). Unfortunately, many factors, both complex and unidentified, contribute to the pathophysiology of mtDNA diseases. For example, the absence of any definitive consistency between a specific genotype and disease phenotype relationship is responsible for instances in which an identical mutation produces vastly different phenotypes between two individuals, or, the same phenotype is caused by different mutations (Morgan-Hughes *et al*., 1995). Additional factors, such as incomplete penetrance (Macmillan *et al*., 1993), population bottlenecks (Jacobs *et al*., 2006), differential segregation (Brown *et al*., 2001), and tissue-specific selection pres-

Table 1. Benefits and Shortcomings of Mitochondrial Gene Therapy Strategies

Treatment strategy	*Strengths*	*Limitations*
Selective inhibition	• Can target mutant mtDNA with high specificity (RE) • Is potentially adaptable and can be made mutation specific (PNA) • Import has been demonstrated for both PNA and RE	• Limited target mutations (RE)—mutation must produce a unique restriction site • Insufficient binding *in situ* (PNA) • Is only heteroplasmic specific—it is not compatible with homoplasmic disorders
Recombinant mtDNA	• Can treat both heteroplasmic and homoplasmic disorders • Is potentially adaptable, as can be made gene and disorder specific • With improved delivery techniques, it may be possible to deliver whole, functional mtDNA genomes	• Limited mitochondrial import of large DNA constructs • There are no recombinant human mtDNA constructs available • It is likely that large administrative doses will be required to ensure sufficient delivery to enough mitochondria to be beneficial
Allotopic expression	• Can treat both homoplasmic and heteroplasmic disorders • Exploits established nuclear gene delivery techniques • Nuclear-expressed products are likely to reach all mitochondria within a transformed cell	• It is difficult to import highly hydrophobic nuclear-expressed proteins • Requires reconstruction of mtDNA-encoded genes • Limited knowledge of targeting compatibility between MTS and reconstructed mtDNA genes

Abbreviations: MTS, mitochondrial targeting sequences; mtDNA, mitochondrial DNA; PNA, peptide nucleic acid; RE, restriction endonuclease.

sures (Jenuth *et al*., 1997), render mtDNA disorders initially difficult to diagnose in terms of identifying the specific mutation responsible, and ultimately, extremely challenging to treat as individual disorders. It is, however, becoming increasingly evident that at the molecular level, susceptibility to acquiring pathogenic mitochondrial mutations is compounded by further levels of complexity, as compared with defects of the nuclear genome. Such characteristics include sequence composition and density, interactive associative proteins, residency, and DNA maintenance including repair mechanisms, each of which is likely to contribute up to a 10-fold greater incidence of mtDNA mutations (Richter *et al*., 1988; Mecocci *et al*., 1993; Barja and Herrero, 2000), and hence subsequent potential to cause disease.

Molecular interactions of mtDNA

In contrast to nuclear DNA, which consists of approximately 1% mRNA-encoded DNA, mtDNA contains no introns, and, excluding the small noncoding control region (D-loop), is virtually entirely composed of RNA-encoding sequence (Anderson *et al*., 1981). Any mtDNA mutation therefore has a significantly greater probability than nuclear mutations to affect essential sequence that encodes a transcript. The susceptibility to acquiring mutations is exacerbated by the residence of mtDNA within the mitochondrial matrix. Traditionally, mtDNA has been thought to lack protective proteins, as opposed to nuclear DNA with its ordered, tight association with histones; however, this opinion is beginning to change. mtDNA is now known to be in association with transcriptional factors and other DNA-binding proteins (Bogenhagen *et al*., 2003; Garrido *et al*., 2003; Legros *et al*., 2004; Wang and Bogenhagen, 2006) within complexes denoted as *nucleoids*. Evidence to date does, however, suggest that mtDNA complexation is facilitated by potentially random packaging via DNA-bending proteins such as TFAM (transcription factor A, mitochondrial) (Kaufman *et al*., 2007; Kucej and Butow, 2007), which may not offer similar protection as nuclear histones do. Nevertheless, mtDNA is more susceptible to pathogenic mutations because of the close proximity of reactive oxygen species (ROS) that are generated as by-products of OXPHOS of the respiratory chain (Kirkinezos and Moraes, 2001), which is ultimately compounded by less stringent DNA repair mechanisms (Yakes and Van Houten, 1997; de Souza-Pinto *et al*., 2008). This vulnerability is further intensified in mtDNA disease, whereby faulty OXPHOS is thought to result in the increased production of ROS (Esposito *et al*., 1999; Vives-Bauza *et al*., 2006), which in turn results in a greater incidence of mutation.

mtDNA heteroplasmy

Depending on its ATP demand, a cell can contain hundreds of mitochondria, and, as each mitochondrion contains multiple copies of mtDNA, potentially thousands of mtDNA genomes can be present per cell. Within "healthy" cells, mtDNA is generally considered to be homoplasmic in nature, such that mtDNA sequences are essentially identical, and have the same potential OXPHOS capacity (although this "healthy" homoplasmic state is, of course, a conceptual term, as mtDNA accumulates random sporadic mutations over time, these mutations are typically undetectable and hence inconsequential). Mitochondrial heteroplasmy, as opposed to homoplasmy, describes the significant presence of both mutant and wild-type mtDNA within the same cell or tissue (Holt *et al*., 1988). Heteroplasmy is often a critical factor driving mitochondrial disease, in particular those caused by inherited point mutations. The extent of heteroplasmy can range between 0 and 100% (100% mutant mtDNA denotes mutant homoplasmy), because of the thousands of copies of mtDNA per cell, and hence a variable proportion of functional-to-mutant mtDNA exists. Complementation of wild-type mtDNA over mutant mtDNA therefore forms a critical threshold between functional and defective mtDNA that, if crossed, produces a particular disease phenotype (Schon *et al*., 1997). Although typically within the range of 60–90% mutant mtDNA present (Gardner *et al*., 2007), the exact threshold of any given disorder is highly dependent on the particular mutation, but in addition, on the specific cell or tissue containing the mutation and its dependency on aerobic metabolism. It is certain that heteroplasmic threshold effects are indicative of the critical balance between cellular ATP supply and demand; the correlation between mtDNA mutant threshold and ATP expenditure is, however, poorly understood, and requires further investigation.

Targeting Therapeutic Molecules to Mitochondria

Genetic intervention to treat mitochondrial disorders resulting from defective OXPHOS offers an alternative approach to the therapeutic strategies that are currently applied. The use of such an approach ultimately requires efficient delivery of therapeutic molecules to, and within, mitochondria. Although there are many reports of promising *in vitro* proof-of-principle concepts, the conversion of such strategies to treat *in vivo* mitochondria and associated disorders has had mixed success. Nevertheless, almost every strategy exploits mitochondriotropic molecules that facilitate delivery to, and import into, mitochondria via ATP-dependent or -independent mechanisms. The use of such mitochondriotropic molecules is discussed in subsequent sections of this paper.

Mitochondrial targeting sequences

Only 13 of the approximate 1500 mitochondrial proteins are encoded by the mitochondrial genome. The remaining proteins are encoded by the nuclear genome, transcribed within the nucleus, translated on cytoplasmic ribosomes, and actively transported to the mitochondria. These proteins contain "molecular addresses": mitochondrial targeting sequences (MTSs) that direct trafficking to, and facilitate import into, mitochondria. Of the proteins destined for the mitochondrial matrix, the majority contain N-terminal cleavable signals that are extensions of the precursor protein. These signals range from 10 to 80 amino acids in length, are rich in hydroxylated and positively charged residues, lack negatively charged residues, and generally have the ability to form amphipathic helices with one hydrophobic face and one positively charged face (Horwich *et al*., 1985b; von Heijne, 1986; Mukhopadhyay *et al*., 2003). Although these properties are relatively generalized and not exclusively attributed to every matrix-destined protein, the sequence variation, and cryptic (or uncharacterized) MTSs among precursors, indicate that there are almost certainly additional properties that have not been described to date. The afore-

mentioned features are, however, believed to contribute in the recognition and translocation of precursor proteins by cytoplasmic chaperones and the import complexes, translocase of the outer membrane (TOM) and translocase of the inner membrane (TIM), into the matrix compartment.

N-terminal targeting signals have been extensively exploited to facilitate mitochondrial localization and subsequent import of a range of nonmitochondrial molecules. Chemically synthesized peptides have been used to investigate mitochondrial import (Pak and Weiner, 1990; Roise, 1992; Lu and Beavis, 1997), and have been conjugated to molecules, including small single- and double-stranded DNA (Vestweber and Schatz, 1989; Seibel *et al*., 1995), peptide nucleic acid (PNA)-coupled oligonucleotides (Flierl *et al*., 2003), liposome-encapsulated oligonucleotides (Geromel *et al*., 2001), and quantum dots (Hoshino *et al*., 2004). Mitochondrially targeted expression reporter plasmids have also been developed to investigate the mechanisms of mitochondrial import (Mukhopadhyay *et al*., 2003; Krayl *et al*., 2006), and nonmitochondrial proteins have been successfully delivered in a range of fusion proteins, including green fluorescent protein (GFP) (Kanazawa *et al*., 1997; Yano *et al*., 1997; Ni *et al*., 1999; Del Gaizo and Payne, 2003) and the cytosolic enzyme dihydrofolate reductase (DHFR) (Horwich *et al*., 1985a).

The use of MTSs for the delivery of therapeutic molecules as both "synthetic" peptides and within fusion protein expression vectors is discussed in detail in the following sections. Surprisingly, there is a severe deficit of knowledge regarding the efficacy of known and employed targeting sequences, in terms of comparative quantitative analysis among the sequences investigated. Much of the data presented is qualitative, and often a particular approach does not explore more that a single targeting sequence. Such information would be invaluable, especially if particular sequences could confer superior targeting properties over other sequences for specific strategies.

Amphiphilic cationic compounds

Amphiphilic cationic compounds with delocalized charge centers have long been known to accumulate within mitochondria. Examples of this class of molecule are rhodamine 123 (Chen *et al*., 1982) and ethidium bromide (Nass, 1970), two of the earliest and most commonly used mitochondriotropic molecules employed to visualize mitochondria and to deplete mtDNA, respectively. Their delocalized charge centers facilitate entry into the cell and accumulation within mitochondria, translocating both plasma and mitochondrial membranes in response to their membrane potentials. Select mitochondriotropic molecules have been employed to facilitate the delivery of molecular cargo to and within mitochondria, of which key molecules that have therapeutic potential are discussed subsequently.

DQAsomes. DQAsomes are considered to be the most promising candidate for the delivery of DNA to mitochondria (Weissig *et al*., 2000). Derived from the compound dequalinium, DQAsomes are liposomal-like compartments that are suitable for encapsulation of molecules, and have been observed to migrate and localize with mitochondria within living cells. DQAsomes have been demonstrated to bind and condense plasmid DNA (pDNA) (Weissig *et al*., 1998), protect pDNA from DNases (Lasch *et al*., 1999), and deliver pDNA to mitochondria (Weissig and Torchilin, 2000; D'Souza *et al*., 2003), where they dissociate as a result of individual dequalinium molecules preferentially entering mitochondria in response to the membrane potential (Weissig *et al*., 2006). DQAsomes have been employed to mediate delivery of both oligonucleotides and plasmid conjugated to an MTS, to facilitate DNA mitochondrial import. Although colocalization with a mitochondrial-specific dye was observed, the MTS was not sufficient to facilitate DNA internalization into the matrix space (D'Souza *et al*., 2005). It is possible that DQAsomes, or derivatives thereof, may have the potential to facilitate the delivery of larger DNA vectors, such as recombinant exogenous molecules, in a mitochondrial therapy (as discussed later), provided that this system can be coupled with an effective import strategy. Nevertheless, the delivery of such recombinant vectors with DQAsomes has not yet been reported in the literature.

Triphenylphosphonium cations. Triphenylphosphonium (TPP) cations have been investigated specifically in the delivery of small bioactive molecules. The antioxidants ubiquinone and vitamin E have been successfully imported *in vivo* into heart, muscle, brain, and liver mitochondria, facilitated by an alkyl-linked TPP cation (Smith *et al*., 2003). Furthermore, a ubiquinol–TPP conjugate significantly decreased heart dysfunction, cell death, and mitochondrial damage in mice subjected to cardiac ischemia–reperfusion injury (Adlam *et al*., 2005). TPP-facilitated delivery of PNAs has also been investigated. Three- to 4-kDa PNA–TPP conjugates have been shown to accumulate several hundred-fold within the mitochondrial matrix of both isolated mitochondria and in cell culture, in a membrane potential-dependent manner (Muratovska *et al*., 2001). The construction and subsequent import of a range of bis-triphenylphosphonium cations has been described (Ross *et al*., 2006). Although uptake into live cells was not achieved, the authors postulated that the simultaneous administration of two monocations that interact postimport should lead to the development of compounds that demonstrate increased delivery and stability capabilities within the cell and mitochondria. In an attempt to deliver larger molecules, conventional liposomes have been converted to become mitochondrial specific by the incorporation of surface-bound TPP (Boddapati *et al*., 2005). Although these molecules were reported to display colocalization with mitochondrial-specific dyes, there are no reports of their effectiveness as a mitochondrial therapeutic delivery vehicle.

Gene Therapy Approaches to Treat mtDNA OXPHOS Disorders

Selective inhibition of mutant mtDNA

It is estimated that two-thirds of all adults with an mtDNA disease carry a inherited pathogenic point mutation (Chinnery, 2002). As discussed previously, mtDNAs harboring point mutations are classed as being either homoplasmic or heteroplasmic, the disease susceptibility being greatly dependent on the variable proportion of mutant to wild-type mtDNA. On the basis of this principle, it has been hypothesized that if the proportion of defective mtDNA is reduced below the disease-specific critical threshold, cells will revert

to a normal, wild-type phenotype (Fig. 1). As mtDNA is constantly dividing under relaxed replication control (independent of cell cycle), even in postmitotic cells, the inhibition of mutant and propagation of wild-type mtDNA is an innovative strategy to reduce heteroplasmic levels within a diseased cell.

Two strategies have been investigated to modulate heteroplasmy by the selectively targeting of mutant mtDNA. The first employs a mitochondrial matrix-targeted restriction endonuclease to discriminate between mtDNA haplotypes that differ by the presence of unique restriction sites. The second approach makes use of complementary antisense molecules to bind selectively and inhibit mutant mtDNA transcription and replication.

Restriction endonuclease selection of mutant mtDNA. Restriction endonuclease selection of specific mtDNA haplotypes was first demonstrated by Srivastava and Moraes (2001), employing a cytochrome *c* oxidase (COX) subunit 8 MTS–*Pst*I endonuclease. Successful mitochondrial import of the fusion protein, and subsequent cleavage of a *Pst*I-containing mtDNA haplotype in the presence of *Pst*I-absent mtDNA, was demonstrated within a range of cell lines, modulating heteroplasmy in favor of the nontargeted mtDNA. A COX4–*Sma*I endonuclease was used to target the T8993G mutation associated with *neuropathy, ataxia, and retinitis pigmentosa* (NARP) (Tanaka *et al*., 2002). Specific elimination of mutant mtDNA in NARP cybrids (cytoplasmic hybrids) was demonstrated, facilitating repopulation of the wild-type mtDNA and subsequent restoration of both intracellular ATP levels and mitochondrial membrane potential comparable to that seen in controls. A transfected COX8–*Apa*LI endonuclease was employed to modulate heteroplasmy both *in vitro* and *in vivo* (Bayona-Bafaluy *et al*., 2005). Despite the fact that a complete shift in heteroplasmy within 2–6 hr was observed in HP202B mouse heteroplasmic hepatocyte cell lines, only a modest shift in mtDNA heteroplasmy in muscle and brain tissue of a mouse model containing two polymorphic mtDNA sequence variants was detected.

Although a restriction endonuclease approach has merit for specific mutations, it is relatively limited in terms of the range of mutations that can be targeted. Not all mutations will produce a restriction site; moreover, if a site is generated, it may not be unique, and hence would not discriminate between wild-type and mutant molecules. Despite this, the use of *Sca*I restriction endonuclease has been explored, which recognizes differentially occurring multiple restriction sites in both haplotypes of mtDNA in heteroplasmic NZB-BALB/c mice (Bacman *et al*., 2007). Although demonstrating shifts in heteroplasmy, a depletion of both mtDNA haplotypes was acknowledged. Despite this, the authors suggest that by tightly controlled and regulated expression, nonspecific mtDNA depletion could be sufficiently minimized.

Even though each of the four studies discussed previously reported on the absence of nuclear localization of mitochondrially targeted restriction endonucleases, it cannot be ruled out that the restriction endonuclease will not have devastating effects on nuclear DNA, if mistargeted to the nucleus.

Antisense inhibition of mutant mtDNA. Whereas a restriction endonuclease strategy is limited to specific mutations, an antisense strategy has the potential to be adapted for any heteroplasmic mtDNA mutation. This approach employs a complementary DNA sequence-specific molecule to bind to

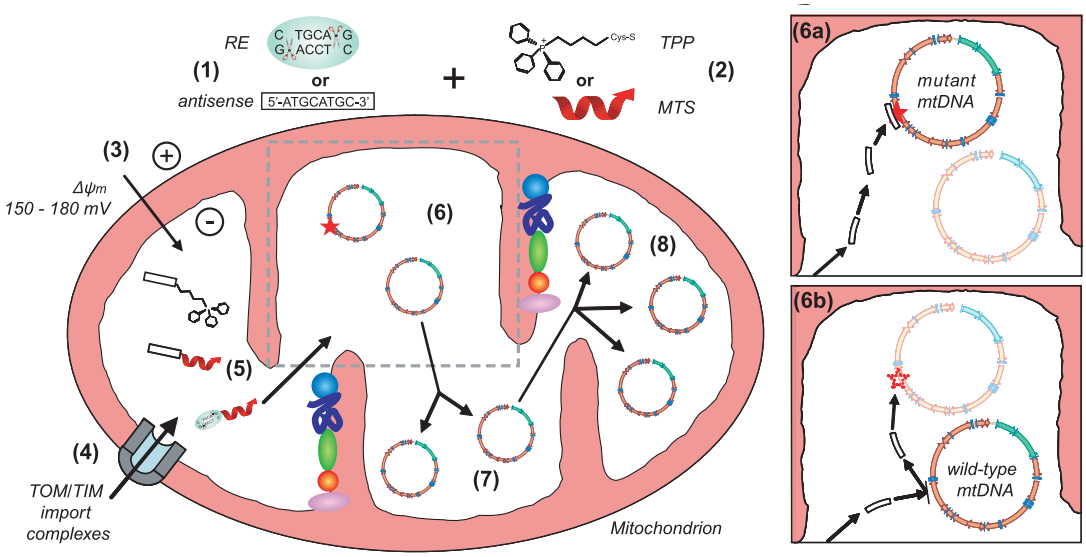

FIG. 1. Selective discrimination and inhibition of mutant mitochondrial genomes. Therapeutic molecules (1) that discriminate between mutant and wild-type mtDNA are conjugated to mitochondriotropic molecules (2), facilitating membrane potential-driven (3) or TOM/TIM-mediated import (4). Targeted molecules accumulate within mitochondria (5) and selectively interact with mtDNA (6) with high specificity for mutant mtDNA (6a), preventing replication of the defective mtDNA and the expression of defective gene products, while not impeding wild-type molecules (6b). Only wild-type, not mutant, mtDNA will continue to replicate (7), and hence will repopulate the cell with functional molecules (8), reducing heteroplasmy below the disease threshold. $\Delta\psi_m$, electrical potential across mitochondrial membrane; mtDNA, mitochondrial DNA; MTS, mitochondrial targeting sequence; RE, restriction endonuclease; TOM/TIM, translocase of the outer membrane/translocase of the inner membrane; TPP, triphenylphosphonium.

its targeted sequence, and inhibit both the replication of the mutant mtDNA and expression of its encoded defective product. Antisense molecules are characteristically single-stranded oligomers composed of DNA, RNA, or chemical analogs, and are typically designed to target and bind their complementary "sense" RNA or DNA target. Although there are reports of DNA oligonucleotide delivery into mitochondria (Vestweber and Schatz, 1989; Seibel *et al*., 1995; Geromel *et al*., 2001; Flierl *et al*., 2003), almost all mtDNA antisense approaches reported use a synthetic DNA-like peptide nucleic acid (PNA), targeting the mtDNA itself. Unlike DNA, PNA nucleotides are connected by a polyamide backbone, rather than by deoxyribose phosphodiester subunits (Nielsen *et al*., 1991). This feature alone enables PNAs to have many characteristics that confer greater functional suitability over many conventional antisense molecules, including a neutral charge, resistance to both nuclease and protease degradation (Demidov *et al*., 1994), and a highly specific hybridization affinity for both complementary DNA and RNA targets (Jensen *et al*., 1997; Efimov *et al*., 1998). The most prominent feature of PNAs, particularly in terms of an mtDNA-specific antisense molecule, is that the presence of a single base mismatch between PNA/DNA hybrids significantly disrupts duplex stability, such that PNAs can selectively discriminate between identical sequences within a heterogeneous DNA mixture that differ only by a single base (Orum *et al*., 1993; Taylor *et al*., 2000). It is this ability that has prompted investigation for the use of PNAs against heteroplasmic mtDNA disorders.

One of the first investigations to use PNAs to target a mitochondrial point mutation was by Taylor and colleagues (1997). 11-Mer PNAs were used to target the A8344G mutation responsible for the disorder *myoclonus, epilepsy and ragged red fibers* (MERRF). Effective binding and replication inhibition of the mutant sequence by a complementary PNA was demonstrated by polymerase chain reaction (PCR)-based replication runoff assays, while maintaining replication competence of the wild-type sequence. Mitochondrial import of PNAs has been investigated, facilitated by conjugation to both mitochondrial-targeting sequences and lipophilic cations. COX8–MTS successfully mediated ATP- and membrane potential-dependent import of PNAs into isolated mitochondria and colocalization with mitochondria in cell culture as observed by confocal microscopy (Chinnery *et al*., 1999). However, despite the MTS, nonspecific nuclear localization was observed. MTS–PNAs have also been successful as a mitochondrial delivery vector of complexed oligonucleotides. A COX4–PNA conjugate was used to deliver fluorescence- and gold-labeled PNA-complementary oligonucleotides to the matrix of isolated and cell culture mitochondria, detected by confocal and electron microscopy, respectively (Flierl *et al*., 2003).

The use of mitochondrially targeted PNAs in culture was first investigated in an *in vitro* disease model, employing a COX8–PNA 11-mer to treat human myoblasts and transmitochondrial cybrids heteroplasmic for the A8344G MERRF mutation (Taylor *et al*., 2001). Although demonstrating import, PNA-mediated replication inhibition of mutant mtDNA was not achieved within isolated mitochondria, nor was a decrease in mutation load detected in cell culture. The same 11-mer was used in a parallel study, employing a triphenylphosphonium cation–PNA conjugate rather than the MTS peptide (Muratovska *et al*., 2001). Although successful localization within isolated and cell culture mitochondria was demonstrated, selective inhibition of A8344G MERRF mtDNA could be achieved only within isolated mitochondria, and no significant modulation of heteroplasmy within cell culture was observed. Both Taylor and coworkers (2001) and Muratovska and coworkers (2001) surmised that, although localizing within the mitochondrial matrix, PNA molecules did not bind sufficiently to mtDNA to prevent replication, possibly because of displacement by replication or transcriptional complexes as they encounter the bound PNA/mtDNA. In light of this, TPP–PNA–benzophenone derivatives have been developed that enhance PNA:DNA duplex stability, and selectively discriminate and bind irreversibly to their target sequence (Ross *et al*., 2003). There are no reports, however, on the subsequent use of these molecules *in vitro* or *in vivo*.

Although most investigations report negligible nuclear uptake by MTS–PNA conjugates, the very observation of nuclear-localized MTS-conjugated (Chinnery *et al*., 1999) and unconjugated/control PNAs (Taylor *et al*., 2001; Flierl *et al*., 2003) permits the unforeseen occurrence of nonspecific binding of nuclear DNA. This risk, although relatively minor, would be reduced by both careful screening of potential PNA molecules against the nuclear genome and monitoring for the occurrence of detrimental nuclear effects.

The use of such an antisense strategy is likely to encounter further issues once successful inhibition is achieved. There is virtually no information about the fate of both mutant mtDNA–PNA complexes and free PNAs that exist once all available mtDNA molecules are bound. It must be assumed that MTS–PNA, and to a greater extent TPP–PNA, molecules will gradually accumulate until either there are no molecules remaining to import or maximal import capacity has been reached. There are at least two issues that need to be addressed. First, by what mechanism, and during what time frame, are PNAs removed/degraded from the mitochondria? Furthermore, is there a saturation point beyond which excessive import results in mitochondrial toxicity and cell death? As the replication and turnover of mtDNA can range between 0.7 and 30 days (Gross *et al*., 1969), and the generation of mtDNA-free rho-zero (ρ^0) cells can take up to 2 months by ethidium bromide treatment (Pon and Schon, 2001), understanding PNA turnover has implications for both the initial inhibition concentration and potential subsequent and repeated administrations.

The second issue is dependent on the first. If selective inhibition is successful, and there are no mutant mtDNA molecules remaining, what is the fate of the unbound PNAs? Considering that it is likely there will be a high concentration of free PNA molecules, is it possible that they will eventually nonspecifically interact and bind wild-type mtDNA? By following this logical progression further, hypothetically, this situation would ultimately lead to the generation of essentially mtDNA-less cells, as all of the mtDNA will be bound and inhibited, which would further exacerbate the disease phenotype. Although these are obvious questions that need to be addressed, they will remain unanswered until efficient delivery and subsequent effective binding are further refined.

Although an antisense approach has merit for modulating heteroplasmy, there are no reports that investigate antisense

DNA analogs other than PNAs. A range of conventional antisense molecules exists to downregulate nuclear genes, and as such, may have potential within an mtDNA gene therapy strategy. Many of these strategies target the removal of mRNA transcripts, and hence modulate gene expression indirectly. Although the conversion of these nuclear approaches into mitochondrial-specific protocols is limited by many of the same hurdles as other mitochondrial strategies, such as effective import, mRNA targeting may not be effective for mtDNA gene therapy in terms of controlling heteroplasmy. As this approach targets mRNA, which relates to the inhibition of mutant protein expression, and not at the necessary DNA level, which is required to eliminate defective mtDNA, stable continuous treatment would be required. Furthermore, although molecules such as shRNA (short hairpin RNA) and siRNA (small interfering RNA) have been demonstrated to be highly effective in downregulating nuclear-encoded mRNA transcripts, it is not known whether the associated interacting proteins complexes, such as Dicer and RNA-induced silencing complex (RISC), are present within the matrix and hence may not be effective. Nevertheless, further development of this strategy may provide useful insight into the regulation of mtDNA gene expression and OXPHOS complex kinetics, in the same way that antisense is employed for the regulation of nuclear gene expression.

Mitochondrial expression vectors: recombinant mtDNA

Nuclear gene delivery strategies employ a gene-carrying vector, traditionally plasmid-like in origin, which is designed to be expressed from within the nuclear compartment. Such vectors have been used for many years to develop our understanding of cellular processes, and in more recent times have been investigated for their therapeutic potential. Conversion of these vectors to become mitochondrially compatible has had little success to date. The development of a mitochondrial vector and an efficient delivery mechanism would have profound applications for use in the study of basic mitochondrial processes, unleashing a vast scope of investigation into the mechanisms of mitochondrial disease, including potentially therapeutic interventions (Fig. 2).

Construction of recombinant mtDNA. The construction of recombinant mtDNA has, in itself, its own challenges. The 16.5 kb of human mtDNA alone makes standard molecular techniques more cumbersome (compared with routine pDNA molecular manipulations), compounding the difficulty of inserting prokaryotic replication origins and selectable markers, necessary features of traditional plasmids. This hampers the acquisition of sufficient recombinant mtDNA for both initial manipulation and subsequent therapeutic purposes, both essential prerequisites for the development of a mitochondrial vector.

The development of nonhuman recombinant mtDNA has had some, albeit limited, success. An *in vitro* random transposition reaction of an *Escherichia coli* origin of replication and chloramphenicol resistance markers allowed the transformation of *E. coli* with recombinant mouse mtDNA, which was maintained at a low copy number (Yoon and Koob, 2003). However, there is evidence to suggest that the use of bacterial DNA within mammalian-destined vectors results in episomal silencing (Chen *et al.*, 2003, 2004) and induction of inflammation, which can be attributed in part to unmethylated CpG motifs (Krieg, 2000). Although these effects can be reduced by the *in vitro* methylation of these motifs,

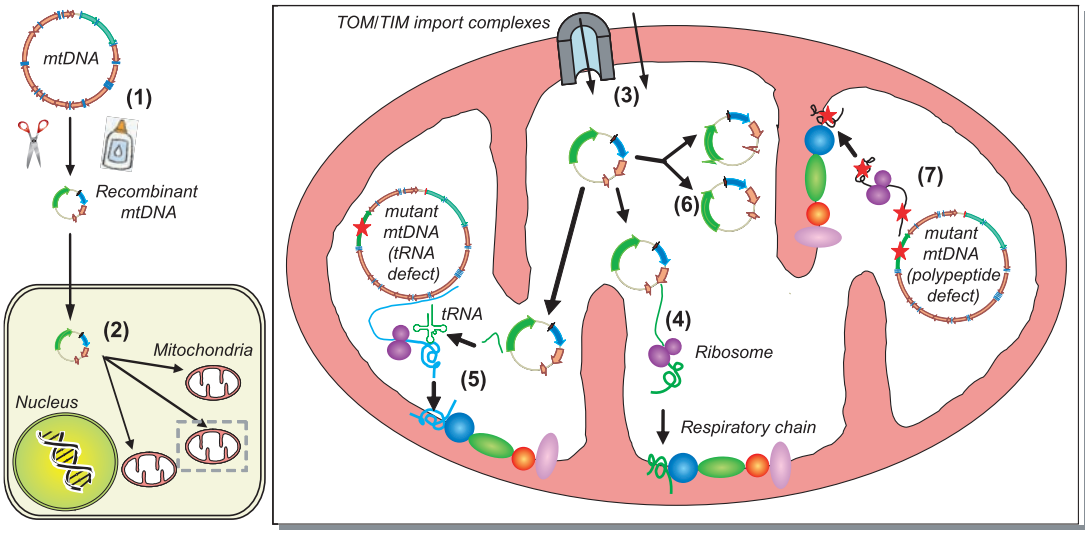

FIG. 2. Design of recombinant mtDNA vectors for stable expression within mitochondria. Critical regulatory and expression components are acquired from human mtDNA, along with the gene of interest, and reduced in size to form a simple mitochondrial expression vector (1). The recombinant vector is delivered into the cell (2), and transported to the mitochondrial matrix (3), facilitated by interactions with mitochondriotropic molecules. Using the transcription and translation machinery, encoded transcripts will incorporate into the inner membrane and OXPHOS respiratory chain (4), or generate tRNAs to be used in the translation of endogenous mtDNA-encoded proteins (5). Stable replication of the recombinant vectors and complementation of new functional genes (6), despite endogenous expression of defective genes (7), should be adequate to reduce the defect load below the disease threshold.

such modifications to CpG islands are commonly associated with gene downregulation. To address this, a bacterial Cre recombinase-mediated excision system has been developed to excise bacterial sequences from recombinant vectors, creating mammalian "minicircle vectors" devoid of bacterial DNA (Bigger *et al*., 2001). To date, there are no reports on the delivery or functional viability of these molecules within mitochondria.

Although there are only a few reports of yeast and some mouse mtDNA genomes modified to facilitate bacterial cloning, there have been no reports of the successful cloning of human mtDNA. Bigger and coworkers (2000) postulated that this difficulty is due to the instability of particular sequences in the human mtDNA genome, previously documented in certain mouse (Kearsey *et al*., 1980) and yeast (Berg *et al*., 1979) DNA sequences when propagated in *E. coli*. Although construction of complete transcriptional and replication-competent human mtDNA genomes would be ideal, it is almost certain that much simpler DNA molecules will be the first vectors tested. These vectors are likely to be considerably smaller than the human mtDNA genome, and may resemble conventional plasmids, that is, constructed of basic components that will facilitate replication, transcription of a multiple cloning site, and having a means for selection. Although the reduction in mtDNA size would be considered in many ways to be advantageous, smaller mtDNA molecules have been reported to have altered characteristics compared with the larger wild-type molecules. mtDNA containing 7.5- and 4.9-kb deletions repopulated 4.8 to 7.5 times faster, respectively, than cells containing wild-type genomes after mtDNA depletion by ethidium bromide (Moraes *et al*., 1999). In addition, a heteroplasmic cell line containing both wild-type and partially deleted mtDNA molecules was rapidly repopulated with the smaller, partially deleted molecule after mtDNA depletion (Diaz *et al*., 2002). Further investigation revealed that heteroplasmic cybrids containing both 4-bp and 7.5-kb deletions preferentially repopulated the smaller mtDNA variant postdepletion. These observations have serious implications for the use of smaller than wild-type recombinant vectors, as they may outcompete functional residing mtDNA, resulting in an acceleration of variable disease characteristics because of an inability to produce endogenous functional OXPHOS polypeptides. Furthermore, translational inhibition of recombinant mtDNA will be exacerbated because of the pool depletion of rRNAs and tRNAs encoded by the wild-type genomes as they are lost. As the exact mechanisms of both copy number control and replication efficiency have not been elucidated, these features will be difficult to control initially in a recombinant DNA vector. Nevertheless, the development of recombinant mtDNA vectors will almost certainly provide invaluable insight into the understanding of these two mechanisms.

Delivery of recombinant mtDNA. Although there has been some, albeit limited, progress in the construction of recombinant mtDNA, the real challenge may actually lie in the delivery itself. To overcome this, a range of protocols has been investigated to introduce exogenous DNA into the mitochondrial matrix. Electroporation was employed to facilitate entry of a 7.2-kb plasmid into isolated mammalian mitochondria, localizing with the inner membrane and within the matrix, without disrupting functional integrity of the mitochondria (Collombet *et al*., 1997). A vector containing a mitochondrially recoded GFP was expressed as a soluble matrix protein and could be detected by fluorometric and immunological analysis, after introduction into yeast mitochondria by microprojectile bombardment (Cohen and Fox, 2001). One of the few examples of uptake within functional mammalian mtDNA was demonstrated by Yoon and Koob (2003). Recombinant mouse mtDNA was electroporated into purified mitochondria from an mtDNA-deficient ρ^0 LL/2 cell line, detecting successful *in organello* RNA expression by RT-PCR. In a unique approach, pDNA was successfully delivered into isolated mitochondria by bacterial conjugation (Yoon and Koob, 2005). By engineering a mitochondrially targeted T7 RNA polymerase, pDNA transfer was subsequently confirmed by the detection of pDNA T7 promoter-mediated transcription. In an interesting alternative to traditional methods, a natural import competence of isolated mammalian mitochondria for DNA has been demonstrated, which could be used as a substrate for DNA synthesis and promoter-driven transcription (Koulintchenko *et al*., 2006). The natural import competence of mitochondria either *in vitro* or *in vivo* has not been reported, although the authors note that in combination with mitochondriotropic molecules, such as DQAsomes, *in vivo* DNA import may be feasible.

To date, mammalian mtDNA delivery is essentially limited to isolated mitochondria; there are limited reports that demonstrate delivery and internalization of functionally competent vectors, and the significant hurdle of *in vitro* and *in vivo* DNA delivery remains. In light of this, these strategies combined with an efficient mechanism to reinsert engineered mitochondria post-DNA delivery may be a suitable approach (Katrangi *et al*., 2007), at least in the development of recombinant mtDNA vectors. Unfortunately, there are no valid reports of recombinant human mtDNA vectors propagated, or delivered, to mitochondria. Although a single report discusses the successful mitochondrial import of a human mtDNA–GFP reporter expression vector by a protein transduction domain (PTD)–MTS–TFAM (mitochondrial transcription factor A) conjugate (Khan and Bennett, 2004), neither their constructs nor their delivery methods have been published in a scholarly manner that would allow scientific peer review and independent confirmation.

Despite the fact that at this time there are no human recombinant mtDNA vectors available, the development of nonhuman vectors should help to maintain progress within this avenue of therapy, and allow the development of delivery protocols to achieve proof-of-principle data in tissue culture and animal models until a suitable human mtDNA-derived vector is available.

Nuclear expression of mitochondrial genes

Mitochondria exist as a product of complementation between the mitochondrial and nuclear genomes. This symbiosis is widely accepted to be the result of the incorporation of an endosymbiont into an ancestral evolving eukaryotic cell more than 1 billion years ago. Mitochondria are now totally dependent on the nuclear genome for the vast majority of its components, having either lost or transferred much of its ancestral DNA content to the nucleus. At least 600 human mtDNA-derived nuclear fragments have been identified (Woischnik and Moraes, 2002); however, this

figure may well be underestimated, as mitochondria import approximately 1500 different proteins that have each acquired targeting signals that allow not only translation on cytoplasmic ribosomes, but importantly, active trafficking across the cytoplasm to their final mitochondrial destination.

The evolutionary migration of mitochondrial genes to the nucleus has ignited interest in the translocation of existing mitochondrial-localized genes to be expressed from the nucleus. Termed *allotopic expression*, mitochondrial genes would be optimized for nuclear expression, and acquire an MTS to facilitate trafficking to and import into mitochondria (Fig. 3). This approach was first explored by Nagley and colleagues (Gearing and Nagley, 1986; Farrell *et al*., 1988; Nagley *et al*., 1988). Mitochondrial import of *in vitro*- and nuclear-expressed mtDNA-encoded proteins fused to an MTS was demonstrated, initially into isolated mitochondria and subsequently into mitochondria within yeast, respectively. Allotopic expression has since been investigated for the treatment of human mitochondrial diseases. The T8993G mutation is responsible for defective *atp6* in two related disorders: NARP and *maternally inherited Leigh syndrome* (MILS). An allotopically expressed *atp6*–COX MTS fusion was successfully imported, processed, and incorporated into the ATPase holoenzyme within T8993G homoplasmic cybrids (Manfredi *et al*., 2002). Furthermore, stable allotopic expression of T8993G cybrids promoted a significant increase in ATP synthesis and growth rate. Guy and colleagues (2002) allotopically expressed a recoded ND4 subunit of complex 1 in an attempt to correct the G11778A point mutation associated with *Lebers hereditary optic neuropathy* (LHON). Treated LHON cybrids demonstrated a 3-fold increase in survival under OXPHOS-selective conditions, and restored ATP synthesis comparable to wild-type cybrids. More recently, allotopically induced oligomycin resistance was established by introducing a variant of *atp6* into both a CHO cell line and NARP cybrids, conferring a significant growth advantage under 1000-fold selection inhibition (Zullo *et al*., 2005).

Not limited to human mtDNA-derived genes, the *Chlamydomonas reinhardtii atp6* nuclear-encoded gene has been successfully expressed within human T8993G cybrids (Ojaimi *et al*., 2002). Remarkably, despite the evolutionary divergence between *C. reinhardtii* and humans (32% conserved homology), algal *atp6* expression enhanced cell viability and ATP synthesis, and although not restored to wild-type levels, the authors note that as with most mtDNA disorders, even slight ATP recovery may overcome the disease threshold to produce a wild-type phenotype. In an alternative approach, human cytoplasmic $tRNA^{Lys}$ was successfully imported into mitochondria *in vitro*, facilitated by a protein complex from the kinetoplastid protozoan *Leishmania* (Mahata *et al*., 2005). $tRNA^{Lys}$ import into mitochondria obtained from patients diagnosed with either MERRF or *Kearns-Sayre Syndrome* (KSS) enhanced translation to nearly wild-type levels, and a reduction of mutant polypeptides was observed.

Despite documented progress, the allotopic expression of mtDNA-encoded polypeptides is still hampered by many limitations. The most widely acknowledged challenge is in overcoming hydropathicity. All human mtDNA-expressed polypeptides localize within the inner mitochondrial membrane, and consequently all contain hydrophobic transmembrane-spanning regions. The extent of hydrophobicity, and an increased susceptibility to hydrophobic interactions, restrict the allotopic potential of these proteins because of a reluctance to be sufficiently unfolded during mitochondrial import, a prerequisite for the translocation of proteins across the TOM and TIM protein import complexes. It has been sug-

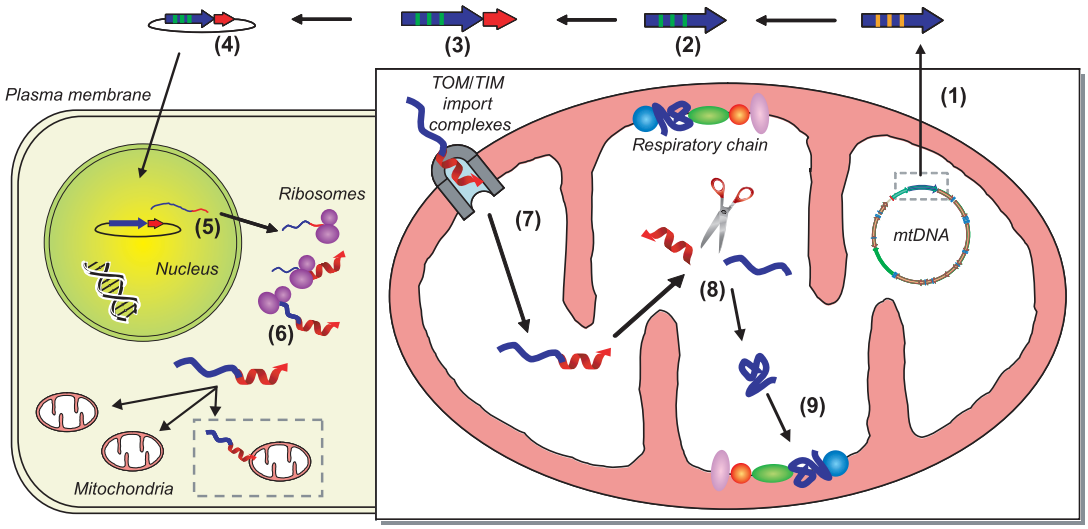

FIG. 3. Allotopic expression of mitochondrial proteins. A mitochondrial gene (1) is recoded for nuclear expression (2) and mitochondrial targeting (3), and inserted into a nuclear expression vector (4). Using nuclear gene delivery techniques, the nuclear localized vector will produce mRNA transcripts of the mitochondrially targeted gene generated by the nuclear transcription machinery (5). The mRNA is transcribed on cytosolic ribosomes, and the protein is actively transported to all mitochondria within the cell (6). Once MTS-facilitated active translocation of the precursor protein through both the TOM and TIM import complexes and into the matrix has taken place (7), posttranslational modifications to the precursor, including cleavage of the MTS, and subsequent folding of the mature protein can proceed (8). The mature, folded, and active protein is incorporated into the inner membrane (9) to perform its functional role with the OXPHOS respiratory chain.

gested that the currently existing 13 genes, at least within human mtDNA, have been conserved because of evolutionary pressure to remain within the mitochondrion itself (Ojaimi *et al*., 2002). Interestingly, of the 13 genes that currently reside within the human mitochondrial genome, only 2 genes, those encoding COX subunit 1 and cytochrome *b*, have been universally retained within their mtDNA by all organisms (Bokori-Brown and Holt, 2006). This suggests that at least the remaining 11 genes have the potential to be relocated to the nucleus. The identification of homologous nuclear-encoded mitochondrial genes from organisms that have already undergone such relocation, and the DNA sequence modifications they have acquired, will provide a benchmark for the translocation of human mtDNA genes. This is evident in the study of *atp6*, of which almost all organisms encode within their mtDNA. The algae *C. reinhardtii* is an exception, whereby its nuclear-encoded *atp6* has acquired a 107-amino acid cleavable MTS, and a lower mean hydropathicity than its mtDNA-encoded homologs (Funes *et al*., 2002). Furthermore, a review of the *reinhardtii* clade revealed an absence of NADH dehydrogenase (*nad*) subunit *3*, *nad4L*, *cox2*, *cox3*, *atp6*, and *atp9* mtDNA genes; each encodes an essential and normally highly hydrophobic component of the OXPHOS respiratory chain (Gonzalez-Halphen *et al*., 2004). It was surmised that reduced hydrophobicity within regions that did not disrupt enzymatic or subunit–subunit interactions was a key prerequisite for nuclear expression. One interesting approach to reduce hydrophobicity has been suggested by de Grey (2000), employing the use of self-splicing protein introns, or *inteins*. This concept aims to separate hydrophobic regions, and thus prevent excessive hydrophobic interactions preimport, while still allowing proteins to regain their original amino acid sequence postexcision. Although this is a theoretical concept, there are examples of organisms that have evolved introns (Daley *et al*., 2002; Gonzalez-Halphen *et al*., 2004) or have undergone fragmentation of a single gene into two subcomponents (Perez-Martinez *et al*., 2001) to express their nuclear-encoded mitochondrial proteins.

Undoubtedly, the modification of existing hydrophobic regions or residues of mitochondrial genes for nuclear expression will be essential to some degree. However, the extent of modification may disrupt the functional characteristics of the protein, and for this reason, has potential limitations. This is evident in an allotopic study of the *atp6* gene, whereby the *C. reinhardtii* nuclear-encoded *atp6* MTS was linked to a human mtDNA-encoded *atp6* gene and expressed from the nucleus (Bokori-Brown and Holt, 2006). Although the algal subunit was targeted to mitochondria more efficiently than human nuclear subunit variants, it was associated with poor survival and growth characteristics during OXPHOS selection, and structural analysis revealed that neither subunit 6 component was incorporated into the ATPase complex.

A further hurdle regarding the delivery of nuclear-encoded mitochondrial proteins is the designation and optimization of targeting signals employed to facilitate delivery. Although there is no consensus regarding the use of specific MTSs used for each and all applications throughout the literature, many studies report the acquisition of unusually long signals by endogenous nuclear-encoded mitochondrial proteins. As opposed to conventional MTSs, which range from 10 to 80 residues, the *C. reinhardtii* nuclear-encoded genes *cox2a*, *cox3*, *atp6*, and *nad4L* (all mtDNA encoded in humans) contain MTSs that are 143, 119, 107, and 133 residues, respectively (Gonzalez-Halphen *et al*., 2004). Interestingly, in two reports, import was enhanced by the duplication of MTS in a tandem repeat (Galanis *et al*., 1991; Oca-Cossioa *et al*., 2003). Such a strategy is not without its limitations, as the duplication of the MTS alone was not sufficient for the import of all of the proteins analyzed. Despite the presence of a long targeting sequence of legume *cox2*, this specific sequence could not be replaced by other targeting sequences (Daley *et al*., 2002). This indicates that a specific relationship between an individual targeting sequence and, at the very least, the N terminus of its mature polypeptide is likely required for successful import (Oca-Cossioa *et al*., 2003), and that the compatibility of unsystematic coupling of targeting sequences and allotopically targeted proteins is potentially limited. This only highlights the importance of further investigation to evaluate a range of artificial leader sequences that differ in both length and combination.

mtDNA and Gene Therapy: Crossing the Threshold of Therapeutic Promise?

Defects of mtDNA are clearly associated with a wide range of human diseases that, at present, have little or no available treatment. Gene therapy offers a practical and potentially efficient strategy to treat the vast array of clinical phenotypes, treating not only the symptoms but the actual cause of the disorder. In light of the fact that many of these disorders are dependent on the heteroplasmic state of their mitochondria and the associated threshold effects, the strategies discussed each have the potential to at least partially decrease the mutation load of a cell, which is in turn likely to have significant influence on the disease phenotype. For mitochondrial gene therapy to be realized and viable in a clinical setting, progress must be made in a number of critical areas of research. The most obvious of these dictates the common theme throughout this review: efficient mitochondrial localization and internalization, and the therapeutic efficacy of the delivered molecule. However, each strategy succumbs to differing levels of complex factors that still must be overcome. Despite their individual merit, both the delivery of recombinant mtDNA and the allotopic expression of mitochondrial proteins are likely to be overshadowed by a selective inhibition strategy to modulate heteroplasmy. This is solely because the delivery of a therapeutic component, be it an antisense molecule or a restriction enzyme, has been relatively well established, and that such components are comparatively small and easily imported, as opposed to the respective delivery of large DNA vectors and hydrophobic proteins. Nevertheless, continued development of these strategies will provide new and invaluable tools to further our understanding of mitochondrial function. In particular, it will shed light on the mechanisms of mtDNA transcription and translation, both of which are important processes that have not been well elucidated and are not fully defined. Although effective treatments for mtDNA disorders are currently unavailable, the rapid progress made to date within these key genetic intervention strategies indicates that the potential for clinical development of at least one of these approaches is real and near, and will be a significant achievement, not if, but when, it is presented.

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